Helix Forge Research Synthetic Sampling Technical Note

HELIX FORGE LAB / p04-full-000161 / RUN R184 / 2026-07-13

ALPHA F1	PRECISION	RECALL	BETA F1
0.873	0.881	0.865	0.846

Decision context

This report records a bounded synthetic sampling evaluation on HF-Eval-S1/S2 v3.2. The 2,400-record population is synthetic. Alpha M-14 is retained as a research checkpoint, not as a deployment authorization.

Engineering interpretation

Aggregate improvement remains meaningful only with the frozen scoring policy. The review therefore keeps slice-level calibration, dataset lineage, and the comparison with Beta B-07 attached to run R184.

Lineage

Upstream evidence: p04-full-000055 (research/papers/2026-07-13/helix-forge-research/failure-analysis/tables/helix-forge-research-failure-analysis-thresholds-v01.csv).

Slice	Records	Precision	Recall	F1
slice-01	227	0.879	0.850	0.864
slice-02	230	0.896	0.863	0.879
slice-03	250	0.878	0.876	0.877
slice-04	184	0.895	0.852	0.873
slice-05	169	0.877	0.865	0.871